

F1 Predictor with LLM Integration

Project Report

1. Introduction

This project focuses on predicting Formula 1 race performance using Machine Learning and Large Language Models (LLMs). The system combines historical Formula 1 race data with AI-powered text generation to create an intelligent race prediction and analysis platform. The project uses FastF1 APIs to collect race data, Scikit-learn for machine learning predictions, and HuggingFace Transformer-based LLMs for generating race insights and commentary. The final system not only predicts race outcomes numerically but also explains predictions in natural language similar to professional Formula 1 analysts.

2. Project Objectives

The primary objective of this project is to create an intelligent Formula 1 race prediction system using Artificial Intelligence techniques. The project aims to: Predict race performance using machine learning algorithms. Analyze driver sector times and qualifying performance. Generate AI-powered race commentary using LLMs. Provide insights into driver strengths and race strategy. Demonstrate the integration of NLP and predictive analytics.

3. Technologies Used

The following technologies and libraries were used in the project: **Python:** Core programming language. **FastF1:** Used for collecting Formula 1 telemetry and race data. **Pandas & NumPy:** Data processing and manipulation. **Scikit-learn:** Machine learning model training. **GradientBoostingRegressor:** Prediction algorithm. **Transformers:** HuggingFace library for LLM integration. **DistilGPT2:** Lightweight language model for text generation.

4. System Workflow

The workflow of the project consists of several stages: **Data Collection:** Historical Formula 1 race data is collected using the FastF1 library. The project specifically uses the 2024 Chinese Grand Prix data.

Data Preprocessing:

Sector times and lap times are converted into numerical values in seconds. Missing values are removed to improve model performance.

Feature Engineering:

The model uses qualifying time and average sector times as features for prediction.

Machine Learning Prediction:

The Gradient Boosting Regressor is trained using historical data to predict future race performance.

LLM Integration:

A Large Language Model generates natural-language race analysis and strategic insights based on the predictions.

Final Output:

The system produces both numerical race predictions and AI-generated commentary.

5. Machine Learning Model

The machine learning component of the project is responsible for predicting driver race performance. The project uses the Gradient Boosting Regressor algorithm because of its high accuracy and ability to model complex relationships. The model is trained using: Qualifying Times Sector 1 Times Sector 2 Times Sector 3 Times The trained model predicts average race lap time for each driver. Drivers are then ranked according to predicted performance.

Feature	Purpose
Qualifying Time	Measures overall driver speed
Sector 1 Time	Early track performance
Sector 2 Time	Mid-track efficiency
Sector 3 Time	Final sector performance

6. LLM Integration

The project integrates Large Language Models (LLMs) to generate professional race analysis. Traditional prediction systems only provide numerical outputs, but this project enhances user interaction by adding natural-language explanations. The HuggingFace Transformers library is used to load the DistilGPT2 language model. The generated prompts include predicted race results, and the LLM creates insights such as: Predicted race winner Strong performing drivers Potential surprises Strategic observations Overall race analysis This integration demonstrates how NLP can improve the usability and intelligence of predictive systems.

7. Advantages of the System

The combination of machine learning and LLMs provides several advantages: Improved user experience through AI commentary. Human-readable race analysis. Advanced sports analytics capabilities. Demonstration of real-world AI integration. Scalable architecture for future improvements.

8. Future Scope

The project can be enhanced further with additional features such as: Real-time race prediction during live events. Weather data integration. Tire strategy optimization. Deep learning models such as LSTM networks. Interactive dashboards using Streamlit. Driver sentiment analysis from interviews and media. Season-long championship simulations.

9. Conclusion

The F1 Predictor with LLM Integration project successfully demonstrates the combination of Machine Learning and Natural Language Processing in sports analytics. The system predicts Formula 1 race performance using historical data and enhances the results using AI-generated commentary. This project highlights the practical use of AI in real-world applications and demonstrates skills in: Machine Learning Data Science Predictive Analytics Natural Language Processing LLM Integration Python Development Overall, the project represents a modern AI-powered sports prediction system capable of both numerical prediction and intelligent analysis.

10. References

FastF1 Documentation Scikit-learn Documentation HuggingFace Transformers Documentation Python Official Documentation Formula 1 Official Statistics